

Invariant Measures on Compatible Classical Histories

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Abstract

The compatible-history program requires a measure on complete trajectories that does not depend on canonical coordinates, temporal slicing, or gauge. For deterministic classical systems, that measure need not be invented as an independent product over time. A physical initial state determines a complete solution, so invariant measure on reduced phase space induces a history measure by pushforward through the solution map. This paper develops that construction and tests its unresolved invariances. A continuous compatibility functional converges under temporal refinement and agrees to 1.8×10^{-5} bits after a nonlinear canonical transformation of an anharmonic oscillator. Reparameterizing the history by linear, quadratic, and exponential clocks gives a cross-clock spread of 5×10^{-6} bits when the lapse is included, while naive slice weighting produces a 0.0814-bit artifact. Finally, a parametrized oscillator with Hamiltonian constraint is evaluated on two nonlinear gauge surfaces. The gauge-fixed Faddeev–Popov measure reproduces the reduced phase volume within 0.0029% and the survival information within 3.6×10^{-5} bits; omitting the determinant creates false shifts as large as 0.254 bits. Surviving-volume information and posterior Kullback–Leibler information are also shown to be distinct. The result supplies a finite constrained classical history measure; its extension to gravity and its proposed relation to cosmological expansion remain open.

From invariant states to invariant histories

The preceding paper, *Noether Constraints and the Measure of Compatible Histories*, replaced coordinate-cell cardinality with Liouville phase volume. Its numerical tests separated two objects sharply. Intersecting-cell counts retained grid shape and canonical coordinates, while Liouville band density agreed across those choices. The paper then asked for the analogous invariant measure on complete constrained histories and for a temporal normalization that survives refinement.

The apparent difficulty is that a path contains a value at every time. If each temporal slice is counted as another independent coordinate, raw history cardinality diverges as the slicing is refined. Deterministic dynamics, however, makes those values dependent. Once physical initial data and the dynamical law are fixed, the trajectory is fixed. The space of solutions is therefore not an infinite Cartesian product of independently chosen states; under suitable existence and uniqueness conditions it is another representation of physical initial-data space.

Let Γ_{phys} be the reduced phase space of physical initial data, after constraints and gauge redundancy have been removed, and let ν_0 be a normalized measure on it. For ordinary Hamiltonian mechanics, one may write

$$d\nu_0(z) = f_0(z) d\mu_L(z), \quad \int_{\Gamma_{\text{phys}}} d\nu_0 = 1, \quad (1)$$

where z denotes initial data, $d\mu_L$ is Liouville measure, and f_0 is a specified ensemble density. Let

$$F : \Gamma_{\text{phys}} \longrightarrow \mathcal{H}, \quad z \longmapsto \gamma_z \quad (2)$$

be the solution map into the set $\mathcal{H}$ of complete classical histories. The induced history measure is the pushforward

$$\boxed{\mu_{\mathcal{H}} = F_*\nu_0}, \quad \mu_{\mathcal{H}}(B) = \nu_0(F^{-1}(B)) \quad (3)$$

for every measurable set of histories $B \subseteq \mathcal{H}$.

Equation (3) is the central construction. It assigns history volume without multiplying coordinate volumes at successive times. Liouville’s theorem preserves the underlying phase volume along Hamiltonian flow, while the solution map records how one physical state is represented as an entire path [1, 2].

The construction also makes canonical invariance direct. Let $T : \Gamma_{\text{phys}} \rightarrow \Gamma'_{\text{phys}}$ be a canonical transformation, let $\hat{T}$ be its action on histories, and suppose the transformed solution map satisfies $F' \circ T = \hat{T} \circ F$. Since $T_*\nu_0$ is the same physical measure in the new coordinates,

$$F'_*T_*\nu_0 = \hat{T}_*F_*\nu_0. \quad (4)$$

Thus both descriptions assign the same measure to the same physical history event. Coordinate invariance is inherited rather than imposed afterward.

CONCEPTUAL PERSUASIONS

- ▶ A deterministic history does not contain one independent state per time slice. It contains one physical initial condition represented along its orbit.
- ▶ The continuum path measure is therefore induced from reduced phase space; it is not obtained by counting increasingly many temporal cells.
- ▶ Canonical invariance of the initial measure propagates to history space through the solution map.
- ▶ The ensemble density f_0 remains physical input. Invariance specifies how an ensemble transforms, not which cosmological ensemble nature chooses.

Compatibility as a functional on histories

Let $Y_{\leq t}$ denote the physical record accumulated through time t . Represent compatibility with that record by a measurable functional

$$0 \leq W_t(\gamma) \leq 1. \quad (5)$$

A hard compatible set $\mathcal{H}_t$ is recovered with $W_t = \mathbf{1}\{\gamma \in \mathcal{H}_t\}$. A soft record can instead grade histories continuously. The surviving history weight and its information are

$$Z_t = \int_{\mathcal{H}} W_t(\gamma) d\mu_{\mathcal{H}}(\gamma), \quad C_{\text{surv}}(t) = -\log_2 Z_t. \quad (6)$$

When $W_{t_2} \leq W_{t_1}$ for $t_2 \geq t_1$, the compatible weights form a filtration and C_{surv} is nondecreasing. This is the measure-theoretic analogue of the original nested-neighborhood theorem.

The same compatibility functional defines a posterior history measure,

$$d\mu_t(\gamma) = \frac{W_t(\gamma)}{Z_t} d\mu_{\mathcal{H}}(\gamma). \quad (7)$$

Its information gain relative to the prior is the Kullback–Leibler divergence [3],

$$\begin{aligned} C_{\text{KL}}(t) &= \frac{1}{\ln 2} D_{\text{KL}}(\mu_t \| \mu_{\mathcal{H}}) \\ &= C_{\text{surv}}(t) + \int_{\mathcal{H}} \log_2 W_t(\gamma) d\mu_t(\gamma). \end{aligned} \quad (8)$$

Because $W_t \leq 1$, the final term is nonpositive and therefore

$$0 \leq C_{\text{KL}}(t) \leq C_{\text{surv}}(t). \quad (9)$$

For a hard indicator, $W_t = 1$ on every surviving history and the two quantities coincide. For soft compatibility they generally differ. Moreover, realized posterior KL information need not be monotone for every observation sequence, whereas nested surviving volume is monotone by construction.

The numerical tests use the dimensionless soft functional

$$W_T(\gamma) = \exp \left[-\frac{1}{2\sigma^2} \int_0^T (x_{\gamma}(t) - y(t))^2 dt \right], \quad (10)$$

where $y(t)$ is the stipulated record and σ fixes its physical resolution. This is a bounded compatibility score, not a claim that a particular observational noise law has been derived.

For a harmonic record $y(t) = \cos t$, $\sigma = 0.35$, and a uniform Liouville ensemble on $(x_0, p_0) \in [-1.5, 1.5]^2$, both information quantities increased over the tested record, but they were not equal.

Table 1: Accumulation of surviving-volume and posterior information. The difference is structural for a soft compatibility functional.

T	C_{surv} (bits)	C_{KL} (bits)	Difference (bits)
0.5	1.558594	0.969143	0.589450
1.0	2.238907	1.309413	0.929494
2.0	3.568521	2.319828	1.248693
3.0	4.195540	2.872346	1.323194
4.0	4.546205	3.169442	1.376764

CONCEPTUAL PERSUASIONS

- “Remaining volume” and “information learned by an observer” are not synonyms once compatibility is soft.
- The original cosmological bridge refers most directly to C_{surv} because it is the negative logarithm of surviving measure.
- Posterior KL information is a complementary observer-relative quantity; it should not be substituted silently for compatible-history volume.
- A physical application must specify the ensemble, record, and resolution before either number is calculated.

Temporal refinement requires the lapse

Let $t = f(\tau)$ be a monotone reparameterization of the same history and define the lapse $N(\tau) = dt/d\tau > 0$. Equation (10) becomes

$$W_T(\gamma) = \exp \left[-\frac{1}{2\sigma^2} \int d\tau N(\tau) (x_\gamma(f(\tau)) - y(f(\tau)))^2 \right]. \quad (11)$$

The lapse is the Jacobian that makes the exponent an integral over physical time. Omitting it assigns equal physical weight to unequal clock intervals and makes the apparent information depend on the reporting parameter.

The harmonic ensemble was evaluated with 320 midpoint slices using linear, quadratic, and exponential clocks over the same physical interval $0 \leq t \leq 4$. With the lapse, the cross-clock spread in C_{surv} was 5.3×10^{-6} bits. Replacing $N(\tau)d\tau$ by a constant slice weight produced a 0.081382-bit spread.

Table 2: Clock reparameterization of one physical history functional. The invariant column includes $dt/d\tau$; the naive column weights coordinate slices equally.

Clock	Invariant C_{surv}	Invariant C_{KL}	Naive C_{surv}
Linear	4.5460702	3.1690652	4.546070
Quadratic	4.5460682	3.1690619	4.464688
Exponential	4.5460649	3.1690584	4.488662

For the quadratic clock, doubling the slice count from 160 to 320 changed C_{surv} by 1.1×10^{-5} bits. The temporal divergence in raw path cardinality is therefore not a required feature of deterministic history measure. It arises when temporal labels are counted rather than integrated with their physical measure.

CONCEPTUAL PERSUASIONS

- Time reparameterization does not require a preferred density of coordinate slices. It requires the invariant element $dt = N d\tau$.
- Refining the numerical clock then improves quadrature instead of adding new physical alternatives.
- This resolves temporal normalization for the tested deparameterizable systems; it does not select a preferred clock in general relativity.

Nonlinear canonical invariance

Exact harmonic flow and a linear canonical shear provide a useful first gate, but they can hide discretization effects. The stronger test uses

$$H(x, p) = \frac{p^2}{2} + \frac{x^2}{2} + \frac{\lambda x^4}{4}, \quad \lambda = 0.5, \quad (12)$$

and the nonlinear canonical transformation

$$Q = x, \quad P = p + \alpha x^3, \quad \alpha = 0.2. \quad (13)$$

Since $dQ \wedge dP = dx \wedge dp$, Liouville area is preserved. The transformed Hamiltonian is

$$H'(Q, P) = \frac{1}{2}(P - \alpha Q^3)^2 + \frac{Q^2}{2} + \frac{\lambda Q^4}{4}. \quad (14)$$

A scrambled Sobol ensemble of 131,072 initial conditions was drawn from $[-1.5, 1.5]^2$. The physical and transformed Hamilton equations were integrated independently with the explicit midpoint method. The record was the high-accuracy anharmonic solution from $(x_0, p_0) = (1, 0)$, and the functional was Eq. (10) with $T = 4$ and $\sigma = 0.35$.

Because a finite-step integrator is not exactly covariant under nonlinear coordinate changes, agreement is expected only in the refinement limit.

Table 3: Nonlinear canonical refinement. The difference between coordinate descriptions decreases toward zero with the integration step.

Δt	$C_{x,p}$	$C_{Q,P}$	$ \Delta C $	$C_{\text{KL},x,p}$
0.1000	4.223811	4.225701	0.001890	2.810564
0.0500	4.214450	4.214817	0.000367	2.802563
0.0250	4.211163	4.211242	0.000079	2.799688
0.0125	4.209709	4.209727	0.000018	2.798383

The finest discrepancy is 1.8×10^{-5} bits. Its decrease under refinement distinguishes continuum invariance from an accidental identity in one coordinate implementation (Fig. 1).

Constraint reduction and nonlinear gauges

Canonical invariance is necessary but not sufficient for a constrained theory. Gauge-related points are different descriptions of one physical history and must not be counted as alternatives. The standard Hamiltonian treatment represents a constrained system on an extended phase space and reduces it by first-class constraints and gauge conditions [4, 5].

The test system is the parametrized unit harmonic oscillator. Its extended coordinates are (x, p, t, p_t) and its first-class constraint is

$$\mathcal{C} = p_t + \frac{x^2 + p^2}{2} = 0. \quad (15)$$

The parameter labeling the orbit is unphysical; physical time t is itself a phase-space coordinate. Complete physical histories can be labeled by the Dirac observables

$$X = x \cos t - p \sin t, \quad P = x \sin t + p \cos t, \quad (16)$$

which are the position and momentum at $t = 0$ and satisfy $\{X, \mathcal{C}\} = \{P, \mathcal{C}\} = 0$.

For one admissible gauge condition $\chi = 0$, the gauge-fixed representation of reduced measure is

$$d\mu_{\text{phys}}^{(\chi)} = \delta(\mathcal{C}) \delta(\chi) |\{\chi, \mathcal{C}\}| dx dp dt dp_t. \quad (17)$$

The determinant is the finite-dimensional Faddeev–Popov factor [6]. It compensates for the angle at which the gauge surface intersects gauge orbits. If $W(X, P)$ is gauge invariant and each orbit intersects the gauge surface once, the normalized ratio

$$Z_\chi = \frac{\int W(X, P) d\mu_{\text{phys}}^{(\chi)}}{\int d\mu_{\text{phys}}^{(\chi)}} \quad (18)$$

is independent of χ . Geometrically, the Faddeev–Popov determinant is the Jacobian from gauge-surface coordinates to the orbit labels (X, P) .

The physical prior was uniform on $(X, P) \in [-1.5, 1.5]^2$, with reduced Liouville area 9. Three gauges were compared:

$$\begin{aligned} \chi_0 &= t, & |\{\chi_0, \mathcal{C}\}| &= 1, \\ \chi_1 &= t + 0.15xp, & |\{\chi_1, \mathcal{C}\}| &= 1 + 0.15(p^2 - x^2), \\ \chi_2 &= t + 0.10x^2, & |\{\chi_2, \mathcal{C}\}| &= 1 + 0.20xp. \end{aligned} \quad (19)$$

The coefficients and bounded domain were chosen so both nonlinear determinants remain positive. Each orbit therefore crosses each test surface once; the experiment deliberately avoids Gribov copies.

Independent scrambled Sobol samples were drawn on each nonlinear gauge surface. A mask selected the common square in Dirac-observable space, and Eq. (10) was evaluated from (X, P) . Estimates were refined from 65,536 to 1,048,576 points.

Table 4: Constrained-history gauge test at 1,048,576 samples. The Faddeev–Popov columns reproduce reduced measure; the last column omits the determinant as a controlled failure.

Gauge	Area	$C_{\text{surv}}^{\text{FP}}$	$C_{\text{KL}}^{\text{FP}}$	$C_{\text{surv}}^{\text{naive}}$
$t = 0$	9.000000	4.546071	3.169067	4.546071
$t + 0.15xp = 0$	9.000177	4.546080	3.169041	4.291667
$t + 0.10x^2 = 0$	9.000260	4.546107	3.169089	4.588782

Across the nonlinear gauges, the maximum Faddeev–Popov-weighted discrepancy was 3.6×10^{-5} bits for survival information and 2.6×10^{-5} bits for KL information. The largest relative error in reduced area was 0.00289%. Omitting the determinant produced false shifts of 0.254404 and 0.042711 bits. Gauge redundancy can therefore manufacture precisely the kind of apparent history information the construction is meant to exclude.

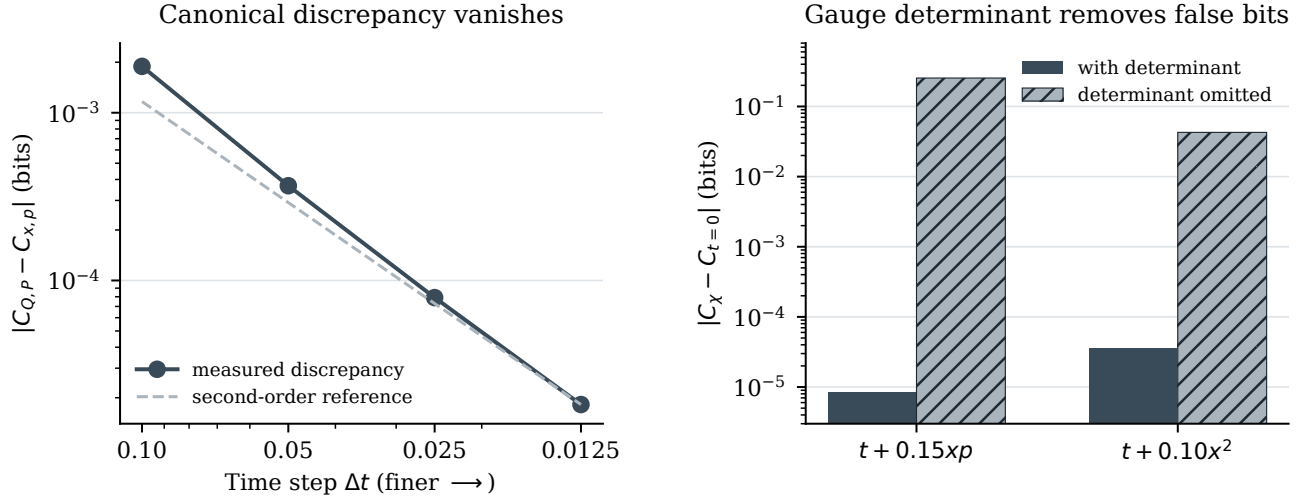


Figure 1: Two continuum tests separate physical history measure from its coordinate representation. Left: independently integrated descriptions of the anharmonic system converge under a nonlinear canonical transformation. Right: the gauge-fixed determinant suppresses false information by three to four orders of magnitude relative to naive gauge-surface weighting.

CONCEPTUAL PERSUASIONS

- ▶ A gauge surface is not automatically a uniformly weighted set of physical alternatives. Its intersection density with gauge orbits matters.
- ▶ The Faddeev–Popov determinant is not a quantum decoration in this test; it is the classical Jacobian required to recover reduced Liouville measure.
- ▶ When the determinant is omitted, gauge choice becomes false history information. When it is included, both surviving volume and posterior information recover their reduced-space values.
- ▶ The test excludes local gauge artifacts in a single-constraint system. It does not address multiple constraints, open gauge algebras, or Gribov ambiguities.

The classical history object

The experiments support one measurement hierarchy for finite deterministic constrained systems. Begin with extended phase space, impose the constraints, quotient gauge redundancy, and normalize a physical ensemble on the resulting space. Push that measure through the dynamical solution map. Finally apply a gauge-invariant compatibility functional to complete histories:

$$\begin{aligned}
 d\mu_{\text{phys}} &\longrightarrow \mu_{\mathcal{H}} = F_* \mu_{\text{phys}}, \\
 W_t(\gamma) &\longrightarrow Z_t = \int W_t d\mu_{\mathcal{H}}, \\
 C_{\text{surv}}(t) &= -\log_2 Z_t.
 \end{aligned} \tag{20}$$

In a gauge-fixed representation, $d\mu_{\text{phys}}$ contains the constraints, gauge conditions, and their determinant as in Eq. (17). In a reduced representation it is simply the symplectic volume on physical degrees of freedom. These are two calculations of the same quotient measure.

The construction resolves the temporal-cell problem for this class of systems. A complete history is weighted once through its physical initial data. Successive observations refine W_t ; successive numerical slices only approximate the invariant time integral. No absolute product of coordinate cell widths or time steps is required.

The result is classical. A Lorentzian Feynman integral assigns complex amplitudes to off-shell paths, not a positive volume on classical solutions. Quantum probabilities for spacetime alternatives require interference to be handled, for example through a decoherence functional on coarse-grained histories [7]. That is a distinct extension rather than a replacement for Eq. (3).

What remains open for cosmology

The accumulated-history proposal relates a three-dimensional relational scale to surviving history information through

$$H_{\text{rel}}(\tau) = \frac{\ln 2}{3} \frac{dC_{\text{surv}}}{d\tau}. \quad (21)$$

The present result strengthens the definition of C_{surv} but does not derive Eq. (21). It also does not show that the history-information rate equals the observed Hubble rate. The constitutive identification between inverse compatible-history volume and physical spatial volume remains a cosmological hypothesis.

Three additional structures are required before that comparison is warranted. First, general relativity has Hamiltonian and momentum constraints at every spatial point rather than one deparameterizable constraint. Its gauge group includes spacetime diffeomorphisms, and global reduced phase space is difficult to construct. Second, observations must be relational and causal: the record cannot be defined by coordinate time or coordinate position. Third, the prior measure on gravitational initial data must be specified independently of the expansion history it is intended to predict.

The finite result nevertheless changes the next question. The missing object is no longer an unrestricted “measure over every value at every time.” It is the reduced symplectic measure on gravitational solution space, together with a causal relational compatibility functional. A finite minisuperspace or constrained cosmological perturbation system is the natural next gate. It should be required to pass canonical, clock, gauge, and refinement tests before any information rate is compared with held-out expansion data.

CONCEPTUAL PERSUASIONS

- ▶ The invariant classical history measure now has a constructive form: reduce physical initial data, then push its measure through the dynamics.
- ▶ The lapse removes clock-density artifacts; the gauge determinant removes orbit-counting artifacts; neither chooses the cosmological ensemble.
- ▶ The Hubble relation remains a constitutive hypothesis. No cosmological rate was used to select the measure, gauge, tolerance, or numerical cutoff.
- ▶ Gravity is the next scientific test, while decoherent quantum histories form a separate later branch.

Constructive conclusion

The complexity-neighborhoods program began with a combinatorial filtration: added distinguishing structure removes fixed-radius neighbors. Its cosmological development replaced objects by histories, geometric proximity by conditional dynamical compatibility, and raw state cardinality by invariant Liouville volume. The present paper completes the corresponding classical history step for finite constrained systems.

Physical initial data carry reduced symplectic measure. Deterministic dynamics maps those data to complete histories. Observations weight the histories, the lapse makes temporal accumulation independent of the clock, and gauge fixing with its determinant makes the result independent of the representative on an orbit. The controlled failures show why every element is necessary: finite time slices and gauge surfaces can both create false bits when their Jacobians are ignored.

The resulting object is compact:

$$\text{invariant compatible-history information} = -\log_2 \int_{\Gamma_{\text{phys}}} W_t(\gamma_z) d\nu_0(z). \quad (22)$$

The tested quantity is canonical, clock, and gauge invariant. Whether a gravitational analogue flows at the expansion rate remains open.

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